

BHARATH NAVEEN

Data Science · Machine Learning · LLM Evaluation · Python · SQL · NLP · Docker · AWS

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SUMMARY

Data scientist and machine learning engineer who builds end-to-end systems, from an ambiguous problem to a deployed, containerized service. Built a production multi-model phishing detection pipeline over 500K+ URL and host features with applied NLP, Dockerized and deployed on AWS. Currently evaluating large language model reasoning with Handshake AI, with three years of enterprise data and analytics experience at TCS.

EXPERIENCE

AI Training & Evaluation Contractor | Handshake AI (Contract, Remote) Sep 2026 – Present

- Design fair, expert-level evaluation challenges that test the reasoning of large language models, targeting known reasoning weaknesses and validated through blind testing so the checks never see the answer key.
- Run ablation, sensitivity, and controlled one-variable comparisons to isolate what drives an outcome, and catch and fix leakage in my own test setups.
- Write structured rationales that separate genuine model errors from ambiguous specifications, providing the feedback used to improve model performance.

System Engineer, Data and Analytics | Tata Consultancy Services Apr 2023 – Aug 2024

- Automated SQL querying and data-collection workflows in Python, saving about 3 days of manual effort per reporting cycle while improving data quality and pipeline reliability.
- Built two self-refreshing Power BI dashboards for two Microsoft stakeholders, replacing manual update cycles with live reporting.
- Applied statistical analysis across 300+ test and 180+ defect records per cycle to flag high-risk areas and surface actionable insights for engineering leads.

Associate System Engineer | Tata Consultancy Services Jul 2022 – Apr 2023

- Led two analysts to coordinate delivery and automate workflows, reducing manual data-collection effort and speeding up defect-pattern analysis.

Assistant System Engineer, Trainee | Tata Consultancy Services Apr 2021 – Jun 2022

- Executed structured data-validation at enterprise scale, building the foundation in data workflows and reporting.

KEY PROJECTS

Phishing Detection System | MS Capstone, University of Arizona 2024 – 2025

- Built a production multi-model phishing detection pipeline (LightGBM, XGBoost, Random Forest, Logistic Regression) over 500K+ URL and host features, with live page capture (Playwright), applied NLP for brand-impersonation detection, and an explainable adjudication layer returning safe, suspicious, or phishing verdicts.
- Used domain-aware dataset splitting to eliminate leakage; containerized with Docker, deployed on AWS, and surfaced through a Streamlit dashboard.

VIP Cohort Analytics Platform | University of Arizona 2024 – 2025

- Designed and built a serverless ETL and inference stack: containerized Julia ETL on AWS ECS/Fargate, DynamoDB with query-tuned indexes, a Lambda API, and multi-environment CI/CD, shipped to production with university stakeholders.

BN Motors SQL Data System | Personal Project 2024

- Designed a normalized 26-table MySQL analytics database from the ER model up, with advanced SQL (joins, window functions, CTEs, subqueries) and reusable reporting views for inventory aging, gross-profit attribution, finance penetration, and customer lifetime value.

Loan Default Prediction | Academic Project 2024

- Ran a full supervised-learning workflow on imbalanced credit data, benchmarking five models (Logistic Regression, Decision Tree, Random Forest, SVM, Gradient Boosting) tuned for recall and F1, and surfaced the real risk drivers.

SKILLS

Languages: Python, SQL, R, Julia

ML / AI: scikit-learn, XGBoost, LightGBM, Random Forest, Gradient Boosting, SVM, NLTK, TF-IDF, LLM integration, RAG

Evaluation: LLM and model evaluation, ablation analysis, A/B testing, data leakage prevention, adversarial evaluation (red-teaming)

Data and Viz: pandas, NumPy, Playwright, BeautifulSoup, Streamlit, Plotly, Power BI

Cloud and Infra: AWS (ECS/Fargate, Lambda, DynamoDB, API Gateway, S3, EC2), Docker, Git, CI/CD, MySQL, PostgreSQL

EDUCATION

MS, Information Science (Machine Learning) | University of Arizona | GPA 3.89 Jul 2024 – May 2026

BE, Mechanical Engineering | SEA College of Engineering and Technology 2016 – 2020

Certification | Microsoft Certified: Azure Fundamentals